



REVA Academy for Corporate Excellence

Program: M. Tech/MSc in Artificial Intelligence	Course: Linear Algebra for AI
Batch: 09	Exam: Final
Date: 14-June-2025	Time: 10:00 AM
Duration: 3 Hrs	Marks: 100

Instructions	
1.	This is an open-book exam. However, all the answers must be paraphrased and customized. Any evidence of copying verbatim from any third-party resources or plagiarized content will lead to suspension and other stringent actions. If any identical answers are found, both will be awarded zero.
2.	You may use any tool or software to answer the questions. All the outputs and codebase must be well commented.
3.	All the answers must be written in the answer script template provided by RACE. The template will be available in the LMS with the answer submission link. Along with the main answer script, all other artifacts like screenshots, codebase and any other materials must be submitted. All such artefacts must be named with the corresponding question number.
4.	All the artefacts must be zipped and submitted through the LMS Link. Zipped Folder must be named as 'SRN_Name of the Course'.

Instructions		Answer ONE FULL question from each unit.	Marks	
UNIT - I				
Q1.	a.	Explain the Direct Approach and the SVD (Singular Value Decomposition) Approach for solving Linear Regression problems. Highlight the differences in terms of: • Mathematical formulation • Computational stability • Suitability for ill-conditioned matrices	5	25
	b.	You are provided with a dataset named LinearRegression_HousingData.xlsx that contains the following columns: • Square Footage: Size of the house in square feet • Bedrooms: Number of bedrooms • Price (\$1000s): Selling price of the house in thousands	10	

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		Implement Linear Regression using the Direct (Normal Equation) Approach and SVD and print the regression coefficients. Compare the outputs from both methods and briefly explain: - Whether the coefficients are similar - In what kind of real-world scenarios, the SVD approach is preferred over the normal equation method.		
	c.	What is the Gram-Schmidt process? Outline the steps for orthonormalizing two linearly independent vectors using this method.	10	
OR				
Q2.	a.	Write the steps involved in K-Means clustering	5	25
	b.	By using Numpy, apply clustering on the given bank dataset, visualize the clusters and provide your interpretation on the clusters	10	
	c.	Apply the following transformations on the given trinagle.png individually using transformation matrices - Scaling (e.g., double the size - Rotation (e.g., rotate by 45 degrees) - Shearing - Reflection (about x-axis or y-axis) - Translation (use homogeneous coordinates).	10	
UNIT - II				
Q3.	a.	Define the Moore-Penrose Pseudoinverse of a matrix. Derive the steps involved in calculating the Moore-Penrose inverse. Discuss its advantages in solving linear systems.	5	25
	b.	Explain the need for specialized storage formats for sparse matrices. Given two sparse matrices in COO format: Matrix A: - row indices = [0, 1, 2] - col indices = [1, 0, 2]	10	

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		- values = [4, 5, 6] Matrix B: - row indices = [0, 1, 2] - col indices = [1, 0, 1] - values = [1, 2, 3] Perform the addition A + B and provide the resulting matrix in COO format.		
	c.	Given the following matrix A, compute the Rank, Determinant, and Trace of A. Also check whether A is orthogonal or not. Provide all intermediate steps and reasoning. $A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	10	
OR				
Q4.	a.	Explain the significance of eigenvalues and eigenvectors in AI applications.	5	25
	b.	Compute the eigenvalues and eigenvectors of the following matrix: $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$	10	
	c.	Write the formulas for cosine similarity and Pearson correlation Compute the cosine similarity and Pearson correlation between the vectors “u” and “v”. $u=[1,0,1,0] \quad v=[0,1,0,1]$ Discuss the relationship between cosine similarity and Pearson correlation for the given vectors.	10	
UNIT - III				
Q5.	a.	Solve the system by Gauss Elimination method. $x + y + z = 6$$2x + 3y + z = 14$	10	25

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		$x + 2y + 3z = 14$		
	b.	For the rectangular matrix $A = \begin{bmatrix} 4 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ • Compute $A^T A$ and AA^T • Find the eigenvalues and eigenvectors of: • $A^T A$, denoted as v (right singular vectors) • AA^T, denoted as u(left singular vectors) • Using the above, compute the Singular Value Decomposition (SVD) of matrix A in the form: $A=U\Sigma V^T$	10	
	c.	Compare LU decomposition and QR decomposition. State their applications in AI.	5	
OR				
Q6.	a.	Compute the least square solution of the equations $2x + 10y = 6$ $4x - 4y = 4$ $2x - 2y = -10$	10	25
	b.	Prove that if $A = [\vec{a_1} \ \vec{a_2} \ \vec{a_3}]$ is an orthogonal 3×3 matrix, then $ \vec{a_1} + \vec{a_2} + \vec{a_3} ^2 = 3$	5	
	c.	Solve the following equations by LU Decomposition method. $3x + y - 4z = 0$ $2x + 5y + 6z = 13$ $x - 3y - 8z = -10$	10	
UNIT - IV				
Q7.	a.	Using the MovieLens 100K dataset, build a user-item rating matrix, apply NumPy-based SVD with top 20 singular values to reconstruct the matrix, and recommend one unrated movie for user ID 50 based on the predicted ratings. Also, briefly explain why SVD is effective in collaborative filtering.	10	25

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		Use the following code for data <pre>import pandas as pd # Load data columns = ['user_id', 'item_id', 'rating', 'timestamp'] df = pd.read_csv('https://files.grouplens.org/datasets/movielens/ml-100k/u.data', sep='t', names=columns)</pre>		
	b.	Explain how SOMs can be used for clustering.	5	
	c.	Build and train a Self-Organizing Map (SOM) using a subset of the MNIST dataset. Visualize the resulting SOM grid to show how similar handwritten digits are organized. Interpret how the SOM preserves topological relationships in the input data. Use the following snippet to load the data <pre>from sklearn.datasets import fetch_openml mnist = fetch_openml('mnist_784', version=1) X = np.array(mnist.data) y = np.array(mnist.target, dtype=int)</pre>	10	
	OR			
Q8.	a.	Briefly describe the architecture of an Extreme Learning Machine (ELM). Highlight the role of: • Input weights • Activation function • Closed-form solution for output weights	5	25
	b.	Critically analyze the advantages and limitations of Extreme Learning Machines (ELM) compared to traditional backpropagation-based neural networks.	10	
	c.	Develop a handwritten digit classification system using Extreme Learning Machines (ELM) on the MNIST dataset using NumPy. <pre>from sklearn.datasets import fetch_openml</pre>	10	

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